

Skills Summary

Interpreting Correlation and Least-Squares Regression

Use this reference while completing your worksheet or when studying.

A least-squares line $\hat{y} = a + bx$ is the line that makes the sum of the squared residuals as small as possible. Three facts follow and drive everything below: the residuals sum to zero, the line passes through $(\bar{x}, \bar{y})$, and r^2 measures how much of the variation in y the line accounts for. The hat on $\hat{y}$ always means *predicted*, never observed.

Skill 1 — Predicting from the regression line

Rule: substitute the x -value into $\hat{y} = a + bx$. The slope b is the predicted change in y for a one-unit increase in x ; the intercept a is the predicted y at $x = 0$, which is meaningful only if $x = 0$ is inside the observed range. Always state the prediction with its units.

Example: A study reports $\hat{y} = 20 + 4x$, where x is hours of training and y is a fitness score. Predict the score at 6 hours.

Step 1. A prediction is a substitution into the fitted equation — no data point is needed, and none is being claimed.

Step 2. $\hat{y} = 20 + 4(6) = 20 + 24$.

44 points is the predicted score at 6 hours.

Try it: For $\hat{y} = 15 + 3x$, predict y when $x = 9$.

check: 42

Skill 2 — Residuals: observed minus predicted

Rule: residual $e = y - \hat{y}$, in that order. A **positive** residual means the point sits above the line and the model under-predicted; a **negative** residual means it sits below and the model over-predicted. The residuals of a least-squares fit always sum to zero, so about half are positive.

Example: At $x = 6$ the model above predicts 44 and the person actually scored 50. Find and interpret the residual.

Step 1. The question compares one real observation with the line, so it is a residual — subtract the prediction from the observation, in that order.

Step 2. $e = 50 - 44$, which is positive.

6 — the point is 6 above the line, so the model under-predicted.

Try it: Another person scored 38 where the model predicted 42. Find the residual and say what its sign means.

check: -4, over-predicted

Skill 3 — Reading r and r^2

Rule: r runs from -1 to 1 : its *sign* gives the direction and its *size* the strength of the linear pattern. Squaring gives r^2 , the fraction of the variation in y accounted for by the line; the rest, $1 - r^2$, is unexplained. Because $|r| \leq 1$, squaring always moves the number toward zero, so r^2 can never exceed $|r|$.

Example: Software reports $r = 0.86$. Find r^2 to two decimal places and say what it means.

Step 1. The question asks how much variation the model accounts for, and that is r^2 — so square the correlation rather than reading r directly.

Step 2. $r^2 = (0.86)^2 = 0.7396$, which rounds to two places.

0.74 — about 74 percent of the variation in y is accounted for, and 26 percent is not.

Try it: A different study reports $r = -0.60$. Find r^2 and say what the negative sign told you that r^2 does not.

check: only shows the direction

Skill 4 — When not to trust the model

Rule: two limits no value of r can remove. **Extrapolation:** a line describes only the range of x actually observed; outside it the prediction has no evidence and often no meaning.

Causation: in an observational study a strong correlation can be produced entirely by a lurking third variable. Only random assignment supports a causal claim.

Example: The fitness model $\hat{y} = 20 + 4x$ was fitted for x between 1 and 8 hours, and scores are capped at 100. What does it predict at 40 hours, and why reject it?

Step 1. Checking an extrapolation means computing the prediction and then testing it against what the context allows.

Step 2. $\hat{y} = 20 + 4(40) = 20 + 160$, far past the 100-point ceiling.

180 — impossible, so the model must not be used at 40 hours.

Try it: For $\hat{y} = 15 + 3x$, fitted on x from 2 to 10 and capped at 100, find the prediction at $x = 50$ and say why it is unusable.

check: 165, past the ceiling and far outside the fitted range

(!) Watch out: A high r^2 says the line fits the observed points, not that the model is right. It cannot detect a lurking variable, it cannot justify a prediction outside the observed range, and it does not become causal evidence when the sample gets larger. Write “accounted for by,” never “caused by,” when interpreting r^2 .